

SELF-SERVICED MEDICAL COMMUNITY WITH BLOCKCHAIN(SMC-BC)

A Project-1 Report

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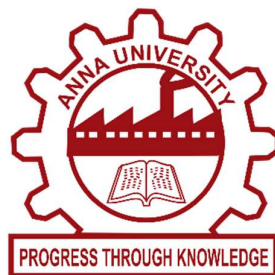
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BONAFIDE CERTIFICATE

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ABSTRACT

A secure medical community is being developed by the Self-Serviced Medical Community with Blockchain (SMC-BC), which incorporates multiple technologies like Federated Learning (FL), Blockchain (BC), and Explainable Artificial Intelligence (XAI). There are still many trustworthy and dependable methods for exchanging health data, however most Electronic Health Record (EHR) data exchanges still take place by mail. The patient's treatment is significantly delayed as a result. By placing the patient at the forefront of the network as well as boosting interoperability and security of health data, blockchain offers the potential to transform health care. The SMC-BC presents a blockchain with permissions system solution for storing and exchanging health information utilizing the Hyperledger fabric framework, as well as introducing a number of principles utilized in Hyperledger Fabric. By employing grant and revocation access procedures, patients have complete control over who can view their medical records. This also highlights the difficulties that must be overcome before Hyperledger Fabric can be successfully used in healthcare systems. Additionally, SMC-BC is combined with XAI to explain the projected disease and with FL to identify the condition while maintaining privacy. SMC-BC illustrates the benefits of utilizing all current technologies for Electronic Health Record Management (EHRM) with one specific application of optical disease management and prediction.

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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
AMD	Age related Macular Degeneration
API	Application Program Interface
BC	Blockchain
CA	Certificate Authority
CA	Certificate Authority
CNN	Convolved Neural Network
EHR	Electronic Health Record
EMR	Electronic Medical Record
FL	Federated Learning
HLF	Hyperledger Fabric
JSON	JavaScript Object Notation
LIME	Local Interpretable Model-Agnostic Explanations
RMSProp	Root Mean Square Propagation
SDK	Software Development Kit
SGD	Stochastic Gradient Descent
SHAP	SHapley Additive exPlanations
SMC	Self-serviced Medical Community
TF	TensorFlow
UI	User Interface
URI	Universal Resource Identifier
WSL	Windows System for Linux
XAI	Explainable Artificial Intelligence

CHAPTER 1

INTRODUCTION

The project, "Self-Serviced Medical Community with Blockchain (SMC-BC)", emerges as a transformative approach. With a strategic integration of cutting-edge technologies like Blockchain, Federated Learning, and Explainable Artificial Intelligence (XAI), we aim to revolutionize Electronic Health Record Management. SMC-BC places patients at the center, providing them unprecedented control over their medical information, all while ensuring security, privacy, and interoperability. As we explore into the comprehensive overview, imagine a paradigm shift where healthcare becomes not just a service but a community-driven, secure, and patient-centric experience.

1.1 OVERVIEW

One of the most important challenges facing the electronic health/medical records (EHR/EMR) environment is ensuring smooth interoperability among various healthcare entities. This difficulty creates obstacles to the efficient interchange of data, which leaves these entities lacking in efficient storage and distribution of data systems. The result is a dispersion of healthcare data that has a negative impact on patient care and overall system performance. Beyond interoperability, data security and privacy are major issues that continue to plague current EHR/EMR systems. Conventional data storage and sharing systems are rife with vulnerabilities that could result in security breaches and weaknesses. The main worry is the enormous risks—including broken security protocols and data leakage—that come with exchanging and keeping medical data. Our initiative, "Self Services Medical Community with Blockchain (SMC-BC)," aims to provide a revolutionary method for managing healthcare data in response to these difficulties. Our initiative uses blockchain technology, namely inside a permissioned framework such as Hyperledger Fabric, to strengthen patient privacy

and medical data security. A strong layer of data protection and access control is incorporated by the use of smart contracts, which guarantee that only authorized parties can access or alter data. Our project's basic idea is that there is an urgent need for a distributed, safe way to exchange and store medical data. Patients have more confidence in the confidentiality and privacy of their data when blockchain concepts are used. Moreover, all entities involved in the healthcare ecosystem can benefit from a holistic view of transactions and interactions, fostering a transparent and collaborative healthcare environment. The SMC-BC project essentially offers a technologically sophisticated and secure platform for healthcare data management in response to the inadequacies of the present EHR/EMR systems. Our project intends to address the complex issues of interoperability, data security, and privacy through the incorporation of Blockchain, Federated Learning, and Explainable AI, ultimately contributing to the emergence of a more robust and patient-centric healthcare ecosystem.

1.2 CHALLENGES FACED IN THE CURRENT SYSTEM

The SMC-BC project essentially offers a technologically sophisticated and secure platform for healthcare data management in response to the inadequacies of the present EHR/EMR systems. Our project intends to address the complex issues of interoperability, data security, and privacy through the integration of blockchain, federated learning, and explainable AI, ultimately contributing to the emergence of a more robust and patient-centric healthcare ecosystem.

1.2.1 Interoperability in Healthcare Data Sharing

The healthcare industry struggles with the requirement for smooth interoperability, especially when it comes to sharing data between various entities. One major challenge is integrating disparate systems, protocols, and standards to enable effective communication. Fostering collaboration between healthcare providers, patients, and other stakeholders requires achieving interoperability. The

development of a unified and patient-centered healthcare ecosystem is hampered by the fragmented nature of healthcare data.

1.2.2 Security and Privacy Concerns in EHR/EMR Systems

There are numerous security and privacy flaws in the conventional ways of storing and exchanging patient data via Electronic Medical Records (EMR) and Electronic Health Records (EHR) systems. Safeguarding sensitive medical data's confidentiality and integrity is a major undertaking. Managing security risks is essential given the growing threat landscape of cyberattacks and data breaches. The project's success depends critically on developing a strong security framework that protects patient data from illegal access, data breaches, and potential vulnerabilities in the system.

1.2.3 Transitioning to a Blockchain-Based Ecosystem

Shifting from conventional data management systems to a blockchain-based model requires a profound understanding of the intricacies involved in deploying and maintaining a permissioned blockchain, such as Hyperledger Fabric. The adoption of blockchain in healthcare signifies a paradigm shift. Overcoming the learning curve associated with blockchain technologies and establishing a seamless transition is crucial for the successful implementation of the SMC-BC project.

1.2.4 Ethical Considerations in Federated Learning and AI Explainability

Integrating Federated Learning and Explainable Artificial Intelligence (XAI) in healthcare introduces ethical considerations related to patient consent, data ownership, and the transparency of AI-driven decisions. Balancing the advantages of advanced technologies with ethical considerations is paramount. Addressing these concerns ensures that the project adheres to ethical standards and garners trust from both healthcare providers and patients.

1.2.5 Overcoming Resistance to Technological Adoption

Resistance to the adoption of new technologies, particularly from healthcare professionals accustomed to traditional methods, poses a significant hurdle. Bridging the gap between legacy systems and innovative technologies is essential for the successful implementation of SMC-BC. This involves addressing concerns, providing comprehensive training, and showcasing the tangible benefits of the proposed technological ecosystem.

1.2.6 Blockchain's Role in Secure Medical Data Management

The integration of blockchain for medical data storage and sharing introduces unique challenges, including establishing trust and ensuring secure access to sensitive healthcare information. Blockchain, while offering decentralized security, requires careful consideration of its application in healthcare. Ensuring that authorized entities access and edit medical details securely is crucial to prevent data leakage and unauthorized access.

1.2.7 AI Integration for Disease Diagnosis and Explainability

Integrating Artificial Intelligence (AI) for disease diagnosis and incorporating Explainable AI (XAI) introduces challenges related to the accuracy of diagnoses, ethical use of patient data, and providing transparent explanations for AI-driven decisions. Harnessing the power of AI in healthcare demands a careful balance between accuracy and ethical considerations. Additionally, ensuring that patients and healthcare providers understand and trust AI-driven diagnoses is critical for widespread acceptance.

1.3 OBJECTIVE

The "Self Services Medical Community with Blockchain (SMC-BC)" project's main goal is to create a secure and robust blockchain network specifically for the healthcare industry. Within the self-serviced medical community, this network is

intended to guarantee the highest levels of security, privacy, and integrity of medical data and transactions. The project seeks to empower individuals as well and healthcare providers by providing them with extensive data management capabilities. Moreover, Federated Learning (FL) integration accomplishes two goals at once: it improves patient data privacy while simultaneously offering researchers access to important data sources. The project develops Explainable AI (XAI) models to envision an interpretable and transparent healthcare environment. These models aim to build trust and an awareness in the medical decision-making process by giving healthcare professionals lucid insights into the logic beneath medical diagnoses and treatment recommendations.

1.4 SCOPE OF SMC-BC

The development of a customized solution, "Self Services Medical Community with Blockchain (SMC-BC)," to address issues with medical data management, with an emphasis on optical diseases, is the main goal of the project. The project aims to create a permissioned, secure blockchain network with Hyperledger Fabric that gives users authority over their health data. The platform allows doctors and patients to use it, guaranteeing safe data management and sharing. Federated Learning is incorporated to protect patient data privacy and to give medical researchers useful information. Explainable AI (XAI) is being applied to optical diseases, producing results that can be easily understood in the way of fundus images that have specific details highlighted. This method serves medical students and provides an invaluable educational resource. The project can be used in practice without requiring specialized hardware because its validity is based on a real-time implementation scenario.

1.5 CONTRIBUTION OF SMC-BC

The "Self Services Medical Community with Blockchain (SMC-BC)" project contributes significantly to healthcare by addressing key challenges. It establishes a

secure blockchain network, ensuring data privacy and integrity. Users gain control over their medical information, fostering a self-serviced medical community. The integration of Federated Learning enhances data privacy while providing valuable insights for medical research. Explainable AI (XAI) models contribute transparent and interpretable results for medical diagnoses, specifically focusing on optical diseases. This practical approach empowers both patients and doctors, facilitating a collaborative and secure healthcare environment. The project's unique contributions make strides in redefining healthcare data management.

1.6 INTRODUCTION TO BLOCKCHAIN

A list of in order arranged blocks makes up the decentralized, reliable distributed ledger known as blockchain, which operates on a peer-to-peer network. Each block forms a chain because it is a hash of the block before it. The genesis block is the first block in the blockchain, and its parent block is the block that came before it. A block has a body and a header. The structure of a Block Header is shown in Table 1.1.

Table 1.1 – Block Header Structure

Version	Specifies the block validation rules.
Previous block hash	This is to ensure that the previous block cannot be changed, without changing the current block header.
Timestamp	The current block creation time.
Merkle root hash	The Merkle root is obtained from the hash values of all the transactions present in the block. This is to ensure that the transactions cannot be modified without changing the header.
Nonce	A nonce is a 4 byte unique number that is used only once in the communication

The core components of blockchain architecture are shown in the Table 1.2.

Table 1.2 – Blockchain Architecture : Components

Node/Peer	A device (like a computer) present inside a blockchain network that has a copy of all the transactions.
Transaction	It is the exchange of information between the two blockchain addresses.
Block	It is used to keep a group of transactions that is distributed among all the nodes in the network.
Chain	A series of blocks arranged in chronological order.
Miners	Before adding any transactions to a new block node, selected nodes carry out the block verification.
Consensus	It is a mechanism through which all of the blockchain network's peers come to an agreement about any transaction in the blockchain.

1.6.1 Permissioned Blockchain

Private blockchain is an additional name for it. There are restrictions on who can participate in transactions and join the network. An organization or business uses a private blockchain for internal purposes. Private blockchains are centralized, meaning that control over the network is held by a single entity. It offers the participants security and transparency. Take Hyperledger Fabric, for instance.

Open and Private: The data is readable by all participants in the blockchain network. Take corporate income statements, for instance.

Closed and Private: The blockchain network is read-only and only accessible to authorized users. For instance, national defense and tax returns.

1.6.2 Hyperledger Fabric

Hyperledger Fabric is a platform for distributed ledger solutions underpinned by a modular architecture delivering high degrees of confidentiality, resiliency, flexibility, and scalability. It is designed to support pluggable implementations of different components and accommodate the complexity and intricacies that exist across the economic ecosystem. It offers some extra features on peak of the blockchain that are ideal for the current situation. Unlike permissionless blockchains like bitcoin and ethereum, this blockchain is closed and permissioned; no one can be added to the network. This resolves the issue of patient data confidentiality. Each organization or common CA in the network has an idea of Certificate Authority called Fabric CA and MSP, that offers identities and checks if a transaction request is submitted. Additionally, the fabric network's components are all pluggable and scalable. It means that an organization can connect to any amount of peers through channels. The fabric offers the idea of smart contracts, just like Ethereum, and when it is packaged and used, it is referred to as chain code. It is a logic that each endorsing peer has implemented. Through Channel Configuration, the channel's founding organizations determine whether to add new organizations. All the transaction data is returned.

1.7 DOCKER PLATFORM USAGE

Docker is an essential part of the "Self Services Medical Community with Blockchain (SMC-BC)" project. It is a popular platform that enables developers to use containers to create, deploy, and manage applications. These software packages, known as containers, are small, lightweight, and self-contained. They include everything needed to run an application and promote consistency between various environments. Docker is essential to the establishment and operation of the network, peers, orderers, and other elements of the Hyperledger Fabric architecture

in SMC-BC. The use of Docker containers promotes uniformity and smooth deployment across a range of environments by guaranteeing the isolation and efficient operation of discrete Fabric network components. The project makes use of Docker Images, which serve as read-only templates that include the configuration files, runtime, libraries, libraries, and application code needed to run the SMC-BC application. Docker files are used to create these images, which outline the steps required to construct an image. In accordance with the principles of Hyperledger Fabric, the project uses Docker Compose files to quickly establish a network. Gaining an understanding of these files is essential to successfully setting up the SMC-BC network and customizing it to the project's requirements.

1.8 INTRODUCTION TO FEDERATED LEARNING

In the SMC-BC project, Federated Learning is a fundamental machine learning technique that allows a centralized model to be trained across various independently connected devices or servers, each of which has local data samples, without requiring direct data exchange. This methodology maintains data privacy and decentralization while facilitating the collaborative development of a shared global model. Federated learning ensures decentralized storage of data on individual devices or servers, solving concerns related to privacy and data security, compared to traditional machine learning where data is collected centrally. The project adopts local model training, which does away with the need to send data to a centralized server for training and instead trains models locally on each device using its own datasets. Only model updates—like gradients or parameters—are sent to a central server or coordinator after local training. After that, these updates are combined to improve the overall model without having direct access to everyone's data, guaranteeing privacy protection. Because federated learning in SMC-BC guarantees that raw data does not leave the individual devices, it is essential to preserving user privacy. Only updated models—which are frequently encrypted or anonymized—are shared, in keeping with the project's dedication to user privacy and data security.

1.9 INTRODUCTION TO EXPLAINABLE ARTIFICIAL INTELLIGENCE

The term "Explainable Artificial Intelligence" (XAI) within the context of our project, Self Services Medical Community with Blockchain (SMC-BC), encompasses the development of AI systems and algorithms designed not only to make predictions or decisions but also to offer understandable explanations for these decisions. This strategic integration of XAI aims to enhance transparency, trust, and accountability within our system. Specifically tailored to the diagnosis of optical diseases, our XAI module employs methods like LIME (Local Interpretable Model-agnostic Explanations) and SHAP (SHapley Additive exPlanations) to determine the key characteristics or inputs influencing the model's output significantly. Notably, this involves highlighting specific areas of fundus images that indicate the presence of disease, rendering the XAI model's output visually informative. Beyond merely improving interpretability, our XAI approach empowers medical professionals and research scholars to confirm and comprehend the decisions made by the AI system. In addition to improving the interpretability of AI-driven results, this method gives medical professionals the ability to confirm and comprehend the decisions made by the AI system. Our Explainable AI module ensures a concrete and approachable way for medical professionals and medical research scholars to understand and validate the diagnostic outcomes by producing outputs that identify areas of concern within medical images. This tangible and approachable method fosters trust and facilitates collaborative decision-making within the broader healthcare community, aligning seamlessly with the goals of SMC-BC.

1.10 ORGANIZATION OF THE THESIS

The organization of this thesis is structured to provide a coherent and systematic presentation of the research conducted in the "Self Services Medical Community with Blockchain (SMC-BC)" project. The chapters are arranged to

ensure logical flow and clarity for the reader. Chapter 1 introduces the project, laying the foundation with an overview, identification of research challenges, objectives, scope, and contributions. It also provides a detailed exploration of the key technologies utilized, including blockchain, Federated Learning (FL), and Explainable Artificial Intelligence (XAI). Chapter 2 presents the Literature Survey, offering insights into existing research papers that inform and contextualize the project. Each subheading within this chapter addresses distinct aspects related to patient-centric health management, electronic health records, consortium blockchain, privacy-focused smart healthcare systems, and privacy-preserving medical data sharing. Additionally, it covers advancements in Explainable AI (XAI) for medical diagnostics and explores decentralized machine learning through Federated Learning (FL) in healthcare. Chapter 3, titled "System Design & Architecture," delves into the specifics of how the SMC-BC system is conceptualized and structured. Chapter 4, "Algorithm Development and Implementation Details," provides a detailed account of the algorithms developed for the project and the practical implementation aspects. Chapter 5, "Results and Discussions," presents the outcomes of the research, facilitating an in-depth understanding of the project's efficacy and limitations. Chapter 6, "Conclusions and Further Work," summarizes the key findings, discusses implications, and proposes avenues for future research and development. The overall organization ensures a methodical exploration of the project, guiding the reader through the research journey from its inception to practical implementation and conclusions.

CHAPTER 2

LITERATURE SURVEY

The literature review includes significant papers that offer complex viewpoints on problems, suggested fixes, and the direction of future research. The project "Self Services Medical Community with Blockchain (SMC-BC)" aims to provide a thorough understanding of the current state of healthcare data management and the guiding principles that inform it, as the insights gained from these research projects come to illumination.

2.1 PATIENT-CENTRIC HEALTH DATA MANAGEMENT

The exploration begins with a patient-centric approach to health data management using blockchain, as proposed in the [1]. The study focuses on empowering individuals to securely manage their health records. The identified challenge revolves around potential data insecurity when sharing information with researchers. An envisioned enhancement involves the integration of an artificial intelligence component to augment the capabilities of the blockchain system . Transitioning into electronic health records (EHR), the [2] introduces a framework utilizing blockchain, Ethereum, and IPFS. This approach combines smart contracts with off-chain storage to address the challenges of transaction speed and gas costs on the Ethereum network, potentially impacting real-time performance. Future directions aim at secure payment integration, improved platform interoperability, and streamlined user interfaces.

2.2 CONSORTIUM BLOCKCHAIN FOR PEER-TO-PEER EMRs MANAGEMENT

A peer-to-peer approach to Electronic Medical Records (EMRs) management and trading is presented in the [3]. The Healthchain system, based on consortium blockchain technology, ensures decentralized data management. Drawbacks include

potential privacy breaches among edge networks and concerns about system security against quantum attacks. Future work encompasses establishing a real-time EMRs data exchange system and implementing a proposed pricing strategic model. The [4] emphasizes privacy preservation in electronic medical record sharing through a blockchain-based smart healthcare system. Identified challenges include security and privacy infringement, along with the complex workflow of the Internet of Medical Things (IoMT). Future work aims to enhance access control principles by simultaneously considering security, privacy, and the intricate IoMT workflow. Delving into privacy-preserving medical data sharing, the [5] introduces MedShare, leveraging blockchain for Electronic Health Record (EHR) sharing with encrypted indexes. Challenges involve potential latency and scalability issues when scaling MedShare. Future work encompasses dynamic access controls, scalability solutions, and integration with healthcare standards for enhanced adaptability and practicality.

2.3 ADVANCEMENTS IN EXPLAINABLE AI(XAI) FOR MEDICAL DIAGNOSTICS

In the evolving landscape of medical diagnostics, a critical exploration goes beyond blockchain applications into the realm of Explainable Artificial Intelligence (XAI). This section delves into recent advancements in methodologies aiming to generate transparent and interpretable results. Understanding the decision-making processes of AI models is pivotal for fostering trust and comprehension among healthcare professionals. XAI not only provides insights into the reasoning behind medical diagnoses but also establishes a bridge between complex algorithms and the end-users, ensuring that the AI-driven outcomes are both reliable and understandable. Recent studies in the field of XAI for medical diagnostics have been instrumental in addressing the "black box" nature of many AI models. These studies focus on developing models that not only produce accurate results but also offer explanations that can be easily interpreted by healthcare practitioners.

Techniques such as attention mechanisms, saliency maps, and rule-based systems are being explored to provide a clear understanding of the features and patterns influencing AI-generated diagnostic outcomes. The overarching goal is to enhance the overall reliability of AI systems in medical diagnostics and promote collaborative decision-making between healthcare professionals and intelligent algorithms.

2.4 FEDERATED LEARNING ADVANTAGES

The exploration into decentralized machine learning approaches, specifically Federated Learning (FL) in healthcare, represents a pivotal aspect of the literature survey. In response to the challenges posed by centralized models and the imperative to maintain data privacy, FL emerges as a promising paradigm. This approach involves training models across distributed edge devices, ensuring that sensitive data remains localized while contributing to the collective learning of a centralized model. Recent research in Federated Learning within the healthcare domain has focused on optimizing the balance between individual data privacy and collaborative model improvement. Studies delve into the development of robust communication protocols, secure aggregation mechanisms, and privacy-preserving algorithms to facilitate effective FL implementation. The investigation also extends to scenarios where data distribution is non-uniform across different edge devices, ensuring the efficacy of FL in diverse healthcare settings. The integration of FL in medical data management holds the potential to revolutionize collaborative machine learning while upholding the paramount importance of patient privacy and data security. This section underscores the significance of decentralized approaches in shaping the future of machine learning applications in healthcare.

2.5 SUMMARY OF THE LITERATURE SURVEY

The comprehensive literature survey spans a diverse spectrum of blockchain applications in healthcare, encompassing pivotal domains such as patient-centric health data management and the establishment of privacy-preserving smart healthcare systems. Within this extensive exploration, challenges and future trajectories have been meticulously identified. These challenges include, but are not limited to, concerns surrounding data insecurity, issues related to transaction speed, and the critical matter of privacy breaches in healthcare data transactions. Furthermore, the survey delves into the realms of Explainable AI (XAI) and Federated Learning (FL) within the healthcare domain. This exploration underscores the growing significance of transparent and decentralized machine learning methodologies, illuminating their crucial roles in the dynamic landscape of medical data management.

2.6 NEED FOR SMC-BC

The literature survey provides a comprehensive understanding of the current state of research and development in the fields of medical communities, blockchain applications in healthcare, explainable AI, and federated learning. This contextualization helps to identify existing systems, frameworks, challenges, gaps and unexplored aspects within the intersection of these domains. It assists in formulating the objective and the need for the SMC-BC and provides a theoretical foundation for the integration of blockchain technology into self-serviced medical communities and the implementation of explainable AI through federated learning. Understanding how previous researchers have approached similar topics helped in shaping the methodology and design of the SMC-BC. This ensures that the SMC-BC contributes meaningfully to the existing body of knowledge in blockchain, explainable AI, and federated learning.

CHAPTER 3

SMC-BC ARCHITECTURE AND DESIGN

The project "Self Serviced Medical Community with Blockchain (SMC-BC)" is explained in this section in terms of its layout. Here, the architectural mechanisms become clear, demonstrating how explainable artificial intelligence, federated learning, and blockchain come together to transform the field of healthcare data management. This section helps to navigate the design concepts and technological interactions that support SMC-BC's strong framework.

3.1 SMC-BC ARCHITECTURE

The architecture above consists of three important submodules namely Hyperledger fabric (HLF), Federated Learning (FL) and Explainable Artificial Intelligence (XAI). The modules are developed independently as they have no dependency during the development. The developed modules are integrated together to obtain the complete proposed methodology of SMC-BC. The architecture diagram of the overall proposed system is shown in Fig. 3.1.

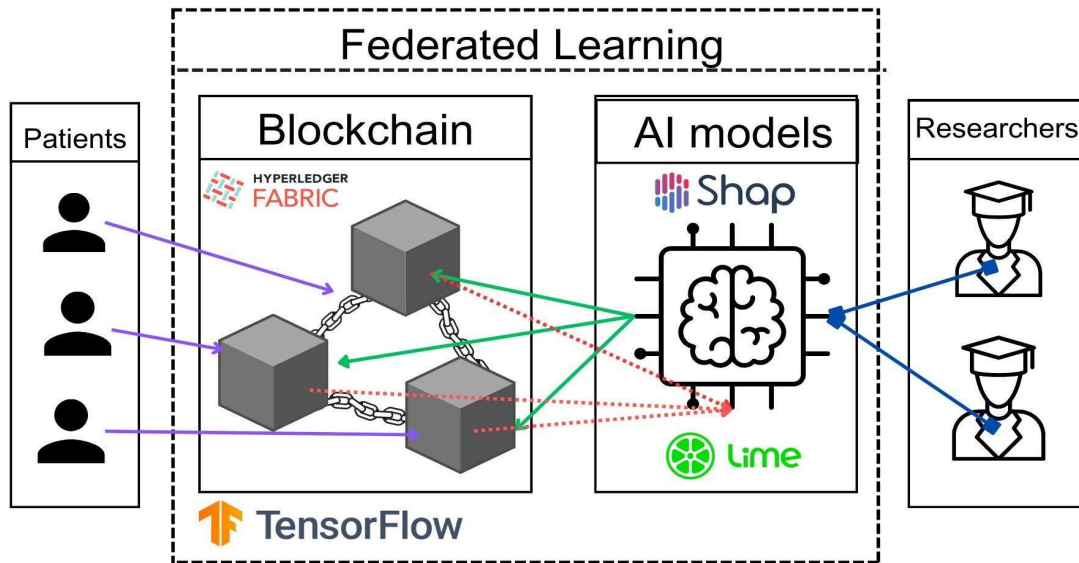


Figure 3.1 – System Architecture

3.1.1 Hyperledger fabric Architecture in SMC-BC

The architecture also provides a detailed color scheme representation of the fabric network. The blockchain operator configures the network's initial configuration and gives the users who run the system the access and credentials they require. The hospital peers are merely image hyperledger/fabric-peer, orderer, and ca docker containers. Since all of the application's components can be added or removed as needed, JavaScript is used to write the backend code and smart contract, and ExpressJS is used as a server to provide REST API. The Angular 11 framework is used in the development of the user interface. REST calls are used for front-end and back-end communication, and JSON web tokens are used for authentication. Additionally, Java, Go, and Typescript—three languages that Hyperledger Fabric officially supports—can be used to write backend code. In the example scenario, two hospital organizations and a single channel called "hospitalChannel" are adequate. It is also possible to successfully add the third hospital in between network connections using the same channel. LevelDB and CouchDB are the two databases that Fabric supports. Because CouchDB is more flexible than LevelDB, it is used in the solution. It is possible to handle images. In contrast to LevelDB, it supports indexes. CouchDB works well because it stores all patient data and does not keep a separate EHR store. While Google created LevelDB, a potent in-memory database for storing key-value pairs and occasionally, it is faster than CouchDB. In the alternative use case scenario, which makes use of hospital EHR databases, references to those records are solely stored in LevelDB as a blockchain ledger (perhaps via APIs). Here, use cases are straightforward, so CouchDB supports it well. The world state and the transaction log are combined to create the ledger. The world state is stored in CouchDB. As a result, for each transaction request, there is no need to query the entire transaction log. Every transaction, beginning with the first one that is kept in the genesis block, is stored in the transaction log.

3.1.2 XAI Architecture in SMC-BC

AI models are developed to classify the images of the eyes of the patient and provide them with an AI predicted report such that the doctor could verify the AI decisions. The ocular dataset is collected and preprocessed by segregating them according to the four classes namely glaucoma, cataract, age related macular degeneration, pathological myopia. The processed data is fed into different model such as the custom Convolutional Neural Network (CNN) and pre-trained models such as MobileNetV2, ResNet50, VGG16. The modular architecture for the Explainable AI model is shown in the Fig. 3.2. The trained model is saved and used for the interpretation of the decisions made using XAI models such as LIME and SHAP.

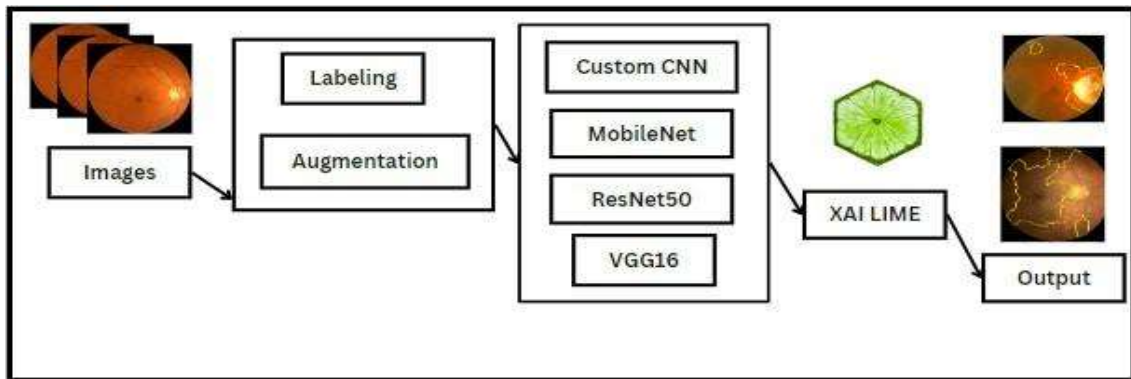


Figure 3.2 – Modular Architecture for XAI

3.2 USE CASES IN SMC-BC

There are three users in the system with role admin, patient, and doctor. Each user has its own use cases. The use case diagram for admin is shown in the Fig. 3.3 , for patients is shown in Fig. 3.4 and for doctors is shown in Fig. 3.5.

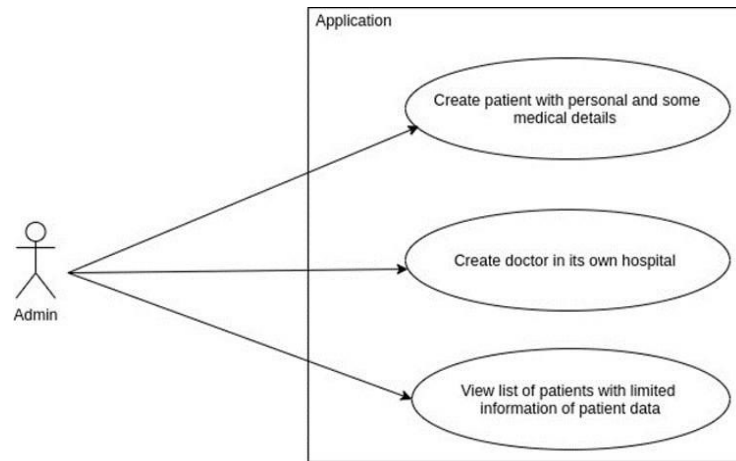


Figure 3.3 – Use Case Diagram for Admin

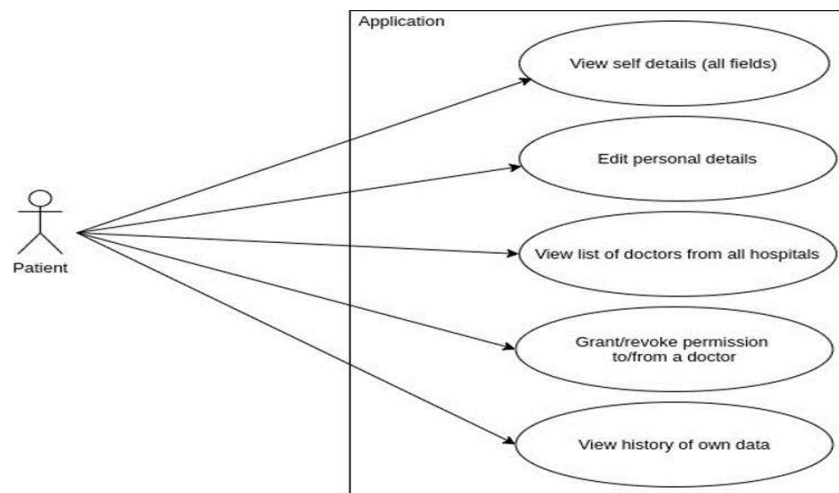


Figure 3.4 – Use case diagram for Patient

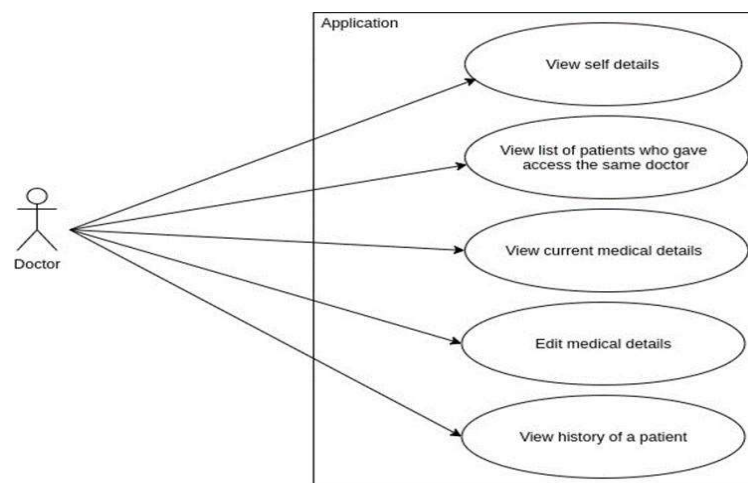


Figure 3.5 – Use case diagram for Doctor

CHAPTER 4

SMC-BC ALGORITHM DEVELOPMENT AND IMPLEMENTATION

This section reveals the lengthy procedure of creating the algorithms that drive SMC-BC's core. The interaction of technological difficulties as the algorithms develop, demonstrating the successful union of federated learning, blockchain, and explainable AI. Here, the story of SMC-BC develops, illustrating how sophisticated algorithmic frameworks for healthcare data management can have revolutionary effects.

4.1 HYPERLEDGER FABRIC ENABLED DATA STORAGE

The HLF comprises many components, has a complex architecture. And these components can be applied and used in many various ways. The following describes how HLF components are developed.

4.1.1 Accessibility with Smart Contracts

A smart contract in hyperledger fabric defines the transaction logic that controls the lifecycle of a business object contained in the world state. Multiple smart contracts are packaged into a single chaincode which is then deployed to a blockchain network. A smart contract defines the rules between different organizations in executable code. Fabric SDK invokes a smart contract to create transactions which then performs changes to the ledger. A smart contract is defined within a chaincode. Multiple smart contracts can be defined within the same chaincode. When a chaincode is deployed, all smart contracts within it are made available to applications. The application comprises mainly three smart contracts packaged into a single chaincode - each role (Patient/Doctor/Admin) invokes its very own smart contract. The hierarchy of the chaincode is shown in the Fig. 4.1.

Admin Contract - This contract is invoked by the admin. The methods in the contract give the admin the ability to create/delete patients by adding/deleting patient objects from the ledger. Admin can also view all the patients available throughout the network.

Patient Contract - The patient interacts with the ledger by invoking this contract. The contract contains the logic that is required for the patient. For instance, only the patient can update/view the personal details and password via the methods defined in the contract. And also the patient contract contains methods to grant and revoke access to from a doctor.

DoctorContract - The doctor contract has methods that allows the doctor to update/read the patient's medical details

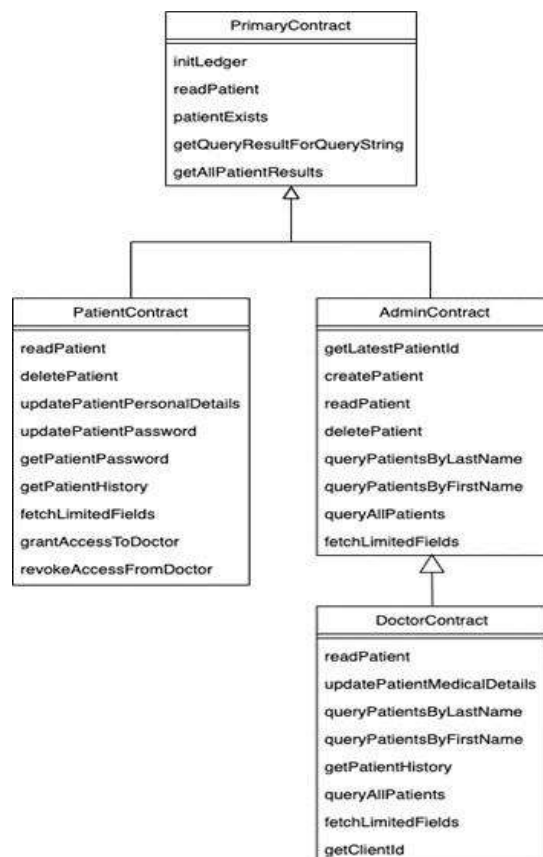


Figure 4.1 – Smart contract / Chaincode hierarchy

The algorithm for initializing the ledger is given below and the notations used are listed in the Table 4.1.

Algorithm for initLedger

Input: None

Output: Initialized ledger with patient data

INITIALIZE ledger → Read 'initPatients' Foreach

patient in JSON file:

Set docType to 'patient' GENERATE

unique key → 'PID' +i STORE patient

record in ledger PRINT “addition of

patient record” END FOR

END

Table 4.1 – Notations Used in the Algorithm

Notations	
Identifier	Explanation
initLedger	Function for Ledger initialization
initPatients	JSON file containing patients data
patient	Predefined class
‘PID’+i	Unique key for each patient
ledger	Blockchain ledger where patient records are stored

4.1.2 Endorsing Policy for Transaction Validation

An endorsement policy for the chaincode in Hyperledger Fabric identifies the peers on the channel that carry out the chaincode operations, endorses the outcomes

to the ledger, and validates the transaction. The peers who will verify and either accept or reject a transaction's execution are specified by the endorsement policies. Peers validate the transaction during this process, and the committing peer makes sure the transaction has the necessary number of endorsements as specified in the endorsement policy.

4.1.3 Data Consistency with Ledger

The world state, which contains the current values of the objects, and the blockchain, which is the history of the transactions leading up to the current world state, make up a distributed ledger in Hyperledger Fabric, which is logically singular but has numerous consistent copies dispersed throughout a network. The concept of a ledger in the application can be compared to a single logical database within a Hyperledger Fabric network. The ledger's format is displayed in Table 4.2. In actuality, the network has multiple copies of a ledger for each hospital's peers, and by consensus, these copies are kept consistent with each other. The applications make use of the ledger in the following ways. First of all, the database containing all of the patient data is the world state. Every patient that is updated and questioned is kept in the global state. The world state is the source of all query transactions. The second type of log file is the transaction log, which is an interconnected block of sequentially structured log files. Each block in the transaction log is made up of a series of transactions, each of which reflects a query or update to the global state. To retrieve an asset's history, `getHistory` API specifically uses the transaction log.

Table 4.2 – Ledger Format

Fields	Description
firstname	Patient's first name
lastname	Patient's last name
age	Patient's age
gender	Patient's gender
phonenum	Patient's contact number
emergencynum	Emergency contact number
address	Patient's address
bloodgroup	Patient's blood group
righteyeimage	Data URI of fundus image of the patient's right eye
righteyepower	Power of the patient's right eye
lefteyeimage	Data URI of fundus image of the patient's left eye
lefteyepower	Power of the patient's left eye
prescription	Prescription given to the patient by the last doctor
symptoms	List of symptoms reported by the patient
riskfactors	Other medical concerns or risk factors
lefteyekeywords	List of medical keywords describing the left eye

righteyekeywords	List of medical keywords describing the right eye
examinationdate	Date of the last eye checkup
changedby	The doctor who last updated the patient's data
permissiongranted	List of hospitals with access to the patient's records
password	Patient's password
pwdTemp	Boolean value; true indicates a temporary password, prompting the user to change it
leftoutput	Data URI of XAI output image for the patient's left eye
rightoutput	Data URI of XAI output image for the patient's right eye
leftclass	Predicted output class of the left eye image (e.g., glaucoma, AMD, cataract)
rightclass	Predicted output class of the right eye image (e.g., glaucoma, AMD, cataract)
report	Health report generated by AI for both eyes

4.1.4 Private Collection of Data

Healthcare uses and stores very sensitive data, which needs to be secure. Only hospitals or physicians to whom the patient has granted access may obtain private patient data. Private data collections can be used to store confidential information using Hyperledger Fabric functionality. Data that needs to be kept secret from other medical facilities or physicians is known as private data. A confidential data collection consists of two parts: the actual private data, which can only be accessed by members of the hospital that are authorized to access private collections of data, and a hash of that data that includes details about patients for which the hospital lacks authorization. The hashed and stored data is inaccessible to hospital members. There are $n! + 1$ private data collection numbers in SMC-BC, where n is the number of hospitals. For example, there will be seven private data collections if there are three hospitals. Every combination will have its own private data collection; for example, each hospital will have its own private collection, any two hospitals can share a private collection, and all hospitals will share one private data collection. The doctors to whom the patient has granted permission will determine which patient data is stored during collection. This will be kept in the hospital's private collection if the patient has only given permission to one hospital. The private data is now transferred from the current collections to the collection that is exclusive to the two hospitals if the patient gives permission for a doctor from another hospital to access it. There won't be any redundant information. Every private data collection will contain patient information at all times. When a patient withdraws their access from a hospital, the same procedures are followed.

4.1.5 Authorization with Fabric SDK

One way to engage with the Hyperledger Fabric blockchain is through the Hyperledger Fabric Client SDK, which offers APIs for interacting with smart contracts, adding transactions to a ledger, and querying the ledger. The following packages are offered by the Fabric SDK:

fabric-ca-client - In order to create trusted identifies on the blockchain network, participants (admins, patients, and doctors) can be enrolled and registered using the fabric-ca APIs. The package establishes a new CA client in order to communicate with the hospital's CA server for participant enrollment and registration.

fabric-common - encapsulates the shared code that all fabric-sdk-node packages use to support fine-grained transaction invocation sending interactions with the Fabric network. offers APIs for event monitoring, logging, environment variable configuration, program arguments, and in-memory settings.

fabric-network - The APIs needed to join the Fabric's network, submit transactions, and view or modify the ledger are included in this package. offers APIs for managing the wallet, which is used to manage identities, and for building connection profiles using the JSON connection profiles that are produced during the creation of a CA.

Within the fabric-network package The Gateway class is the primary class that enables network interaction between the Fabric SDK and the network. After the object is instantiated, it opens a gateway or connection to a peer or user in the blockchain network, providing accessibility to the chaincode and any channels that the peer or user is a member of.

4.1.6 Storing Images in Hyperledger Fabric Block

In the context of the SMC project, the storage of images within a blockchain presents a unique challenge due to the inherent limitations on directly storing binary data in blocks. To address this, a distinctive approach is adopted—converting images into a base64-encoded Data URI format. This conversion is achieved through a specific command:

```
node -e 'console.log(`data:image/png;base64,${fs.readFileSync(process.argv[1], "base64")}`)' /Users/project/image
```

This command takes the path to the image file as an argument and outputs the image content in a base64-encoded Data URI format as shown in the Fig. 4.2, ready for storage within the blockchain. The choice of using a Data URI allows the representation of the image as a string, overcoming the traditional limitations of directly storing binary data in the blockchain.

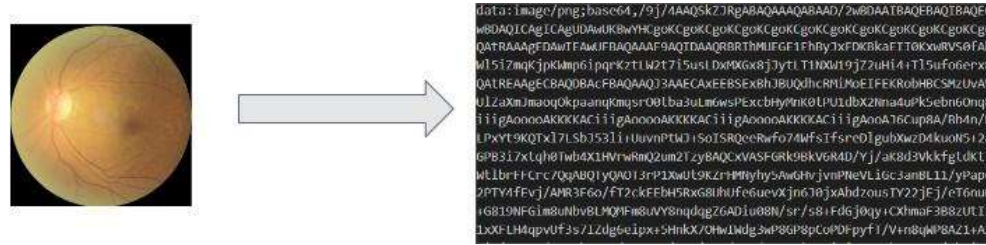


Figure 4.2 – Conversion of Image into dataURI

The images involved in this process include the patient's left and right eye fundus images, as well as the output images generated by Explainable Artificial Intelligence (XAI) for both eyes. The resulting Data URIs serve as compact representations of the images, facilitating their inclusion in the blockchain while adhering to its constraints. This innovative technique ensures the secure and decentralized storage of crucial medical images, maintaining the integrity and privacy of patient data within the SMC-BC blockchain framework.

4.2 FEDERATED LEARNING FOR DECENTRALIZATION PLATFORM

The federated learning is employed with HLF to achieve high security of data while using the patient's medical data to be trained with an AI model. On training the AI models the privacy of the patient should be ensured that none interferes with the communication between the data transfer during the interaction of these two modules. Thus, Federated learning is employed in the HLF network by creating a container of TensorFlow(TF) in Docker and connecting that container to all the peers in the network. The AI model developed is distributed to each peer and the model on each peer gets trained with the help of a TF installed container. The

trained model again aggregates with the global model. This new model is thus used in the prediction. The FL is also triggered when the doctor modifies the class predicted by the AI model to be wrong and feeds the appropriate class that retrain the AI model with that image and newly added class which in turn aggregates to the global model. Thus, employing the FL into the HLF networks integrates the AI model without compromising the security of the patient's data. The TF plays a major role in developing a FL structure inside the HLF network.

4.3 OCULAR DISEASE DATA COLLECTION

Ocular Disease Intelligent Recognition (ODIR) is a structured ophthalmic database of 5,000 patients with age, color fundus photographs from left and right eyes and doctors' diagnostic keywords from doctors. Annotations were labeled by trained human readers with quality control management. They classify patient into eight labels including: Normal (N), Diabetes (D), Glaucoma (G), Cataract (C), Age related Macular Degeneration (A), Hypertension (H), Pathological Myopia (M), Other diseases/abnormalities (O).

4.4 DATA PRE-PROCESSING AND DATA AUGMENTATION

The dataset contains the pool of images without their labeling to their respective class and it also contains a CSV file mapping the images to their classes. From this the required classes and its respective images are categorized with the help of a python script and are stored in their respective folders that are newly created. These data are then classified as train data, test data and validation data. Then data augmentation is performed to artificially increase the size of our dataset by applying various transformations to the existing images. Some augmentation techniques such as rotate, zoom, flip, scale and shear are implemented with the help of image generators as this helps in improving the generalization of our model and reduces overfitting.

4.5 OCULAR DISEASE CLASSIFICATION MODEL DESIGN

In order to classify the image preprocessed , CNN models are developed to get trained with the data. Four different CNN models were employed to learn the classification mechanism in ocular disease prediction namely custom CNN and pre-trained models such as MobileNetV2, VGG16, ResNet50. The pre-trained model MobileNetV2 outperformed all the CNN models.

4.6 OCULAR DISEASE CLASSIFICATION MODEL TRAINING

The designed models are trained with a training dataset which has 4 classes of images of each size 224 X 224 X 3. The training of these models are compiled with various optimizers such as Adagrad, Adam, Stochastic Gradient Descent (SGD), Root Mean Square Propagation (RMSProp). The validation dataset is used to validate the learning of the model during the training process. The trained model is tested against the test dataset and various parameters such as precision, recall, f1-score, support.

4.7 INTERPRETATION WITH XAI INTEGRATION

The XAI packages interpret the decision made by the AI models by getting trained with the data along with the input data. LIME is a powerful XAI tool to highlight the region in the image that contributes to the class the AI model has predicted. The LIME model returns the markers to the highlighted region in the image. This marker is added to the original image such that it explains the area of the image.

CHAPTER 5

RESULTS AND DISCUSSIONS

The section presents the quantitative results derived from the deployment of the "Self Serviced Medical Community with Blockchain (SMC-BC)." The conversation that follows examines these results and offers a thoughtful examination of their consequences. This section provides a framework for in-depth analysis, where the results are examined and talking develops to understand their importance and possible improvements, forming the plot of SMC-BC.

5.1 IMPLEMENTATION ENVIRONMENT

The environment to implement HLF requires a Linux operating system which is enabled by Windows System for Linux (WSL). The docker setup is made to isolate the elements of the HLF. The docker containers run their job in separate ports and contribute to the HLF network. The docker containers that have organization, orderer, couchDB running independently network are made up initially to establish communication between them. Once the networks are up, the channel 'hospitalchannel'. The channel is used to deploy the chain codes in all the organizations and its peers. The peers endorse the deployment of the chain code and create a global ledger for the channel. The ledger consists of a world state that contains the current state and transaction logs which contains the history of the changes made to the system. Once the deployment is successful, the front-end and back-end system are executed to open the web application.

5.2 DATA ACCESSIBILITY IN SMC-BC

A Web Interface is created to make it easier for the users to access the SMC. All the following types of users will have their own interfaces and their respective login credentials.

1. ADMIN

2. PATIENT

3. DOCTOR

5.2.1 Data Accessibility of Admin in SMC-BC

1) Create Patient: A new patient can be created by the admin under the respective hospital by entering all the required details including both personal and medical details of the patient as shown in the Fig. 5.1.

The screenshot shows the 'Create Patient' form within the Titanium Admin interface. The header bar is green with 'Titanium' and 'Admin' on the left, and 'Logout User' on the right. The form title 'Create Patient' is centered, with a 'Refresh' button on the right. The form itself is a white box with a light gray border. It contains a message: 'Please fill out patient details. All fields are required.' followed by several input fields: 'First name' (with placeholder 'Enter your first name.'), 'Last name' (with placeholder 'Enter your first name.'), 'Address' (with placeholder 'Enter your address.'), 'Age' (with placeholder 'Enter your age.'), 'Gender' (with placeholder 'Enter your Gender.'), and 'Blood group' (a dropdown menu with 'Choose your blood group' as the selected option).

Figure 5.1 – Creating a Patient

2) Create Doctor: A new doctor can be added by entering their required details under the respective hospital as shown in the Fig. 5.2.

The screenshot shows the 'Create Doctor' form within the Titanium Admin interface. The header bar is green with 'Titanium' and 'Admin' on the left, and 'Logout User' on the right. The form title 'Create Doctor' is centered, with a 'Refresh' button on the right. The form itself is a white box with a light gray border. It contains a message: 'Please fill out doctor details. All fields are required.' followed by several input fields: 'First name' (with placeholder 'Enter your first name.'), 'Last name' (with placeholder 'Enter your last name.'), 'Hospital' (a dropdown menu with 'Select hospital you belong.' as the selected option), 'Speciality' (with placeholder 'Enter your speciality.'), 'Username' (with placeholder 'Enter doctor's username.'), and 'password' (with placeholder 'Enter doctor's password.').

Figure 5.2 – Creating a Doctor

5.2.2 Data Accessibility of Patient in SMC-BC

- 1) View Patient Details: All the patients can view their own details and their diagnosis report as shown in Fig. 5.3 and Fig 5.4 respectively.

Titanium

Patient

Logout User

Patient PID0

Edit personal details View doctors View history Refresh

Details

Patient ID

PID0

First Name

Jim

Last Name

Andresson

Age

45

gender

Male

Contact number

+93 456-675-3423

Emergency number

+93 345-234-112

Address

123 Main Street, Cityville, State

Blood Group

AB+

lefteyepower

+2.0

righteyepower

+2.5

lefteyekeywords

cataract,macular epiretinal membrane,moderate non proliferative retinopathy,ocular_pressure:18 mmHg

righteyekeywords

normal fundus,drusen,epiretinal membrane,ocular_pressure:16 mmHg

Symptoms

Blurred vision,redness in eyes

Risk factors

History of diabetes

Examination Date

2023-10-05

Prescription

PRINIVIL TABS 20 MG (LISINAPRIL), HUMULIN INJ 70/30 20 units after

Figure 5.3 – Patient Details in UI

AI Diagnosis Report

Left

pathological_myopia

Classification

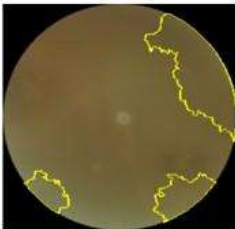
Right

age_related_macular_degeneration

Classification

Left

Diagnosis:



Right

Diagnosis:



Report

Left Eye: highly myopic eyes with degenerative macular changes that present with different patterns and degrees of chorioretinal atrophy and specific findings including lacquer cracks, myopic CNV, and Fuchs' spot. Right Eye: loss

Figure 5.4 – AI Diagnosis Report

2) Edit Patient Details (Personal): A patient can only edit their personal details as shown in Fig. 5.5, while their medical details will be under the control of their respective doctor.

Edit Patient

Please fill out patient details. All fields are required.

First name:

Last name:

Address:

Age:

Gender:

Contact number:

[Refresh](#)

Figure 5.5 – Editing a Patient's Details (Personal)

3) Grant or Revoke Doctors' Access: Every patient can grant access of their respective medical details to the required doctors or can also revoke it back from them after their treatment period as shown in Fig. 5.6.

Doctors

List of all doctors.

Doctor Id	First Name	Last Name	Actions
HOSP2-DOC1	Tamzih	Selvan	GRANT REVOKE
HOSP2-DOC3	Jerry	Mark	GRANT REVOKE
HOSP1-DOC0	Sam	Richard	GRANT REVOKE
HOSP1-DOC2	Raji	Ragavel	GRANT REVOKE

Total count: 4

[Refresh](#)

Figure 5.6 – Managing Doctors' Access

5.2.3 Data Accessibility of Doctor in SMC-BC

- 1) View Enrolled Patients List: Every doctor can view the list of patients enrolled under them and their respective details as shown in Fig. 5.7.

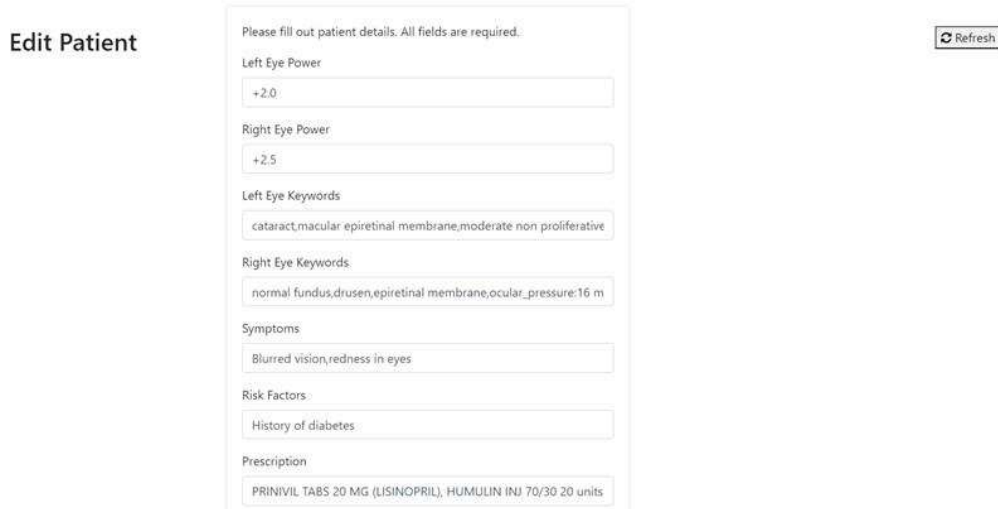


Patient Id	First Name	Last Name	
PID0	Jim	Andresson	View more

Total count: 1

Figure 5.7 – Enrolled Patients List

- 2) Edit Patient Details (Medical): Doctors can edit the medical details of the patient for updating the value fields based on the treatment as shown in the Fig. 5.8.



Please fill out patient details. All fields are required.

Left Eye Power
+2.0

Right Eye Power
+2.5

Left Eye Keywords
cataract, macular epiretinal membrane, moderate non proliferative

Right Eye Keywords
normal fundus, drusen, epiretinal membrane, ocular_pressure:16 m

Symptoms
Blurred vision, redness in eyes

Risk Factors
History of diabetes

Prescription
PRINIVIL TABS 20 MG (LISINAPRIL), HUMULIN INJ 70/30 20 units

Figure 5.8 – Editing a Patient's Details (Medical)

5.3 IMAGE CLASSIFICATION MODEL RESULTS

In this section, the performance of various AI models trained on the image dataset is presented, encompassing Custom CNN, ResNet50, MobileNetV2, VGG16, and an additional custom CNN variant. The accuracy metrics, indicative of the models' predictive capabilities, are summarized in the Table 5.1.

Table 5.1 – Results of Various Models

MODEL	ACCURACY	OPTIMISER
Custom CNN	82%	Adam
ResNet50	78%	RMSProp
MobileNetV2	88%	SGD
VGG16	64%	RMSProp

Additionally, specific attention is given to the MobileNetV2 model's performance concerning ocular disease detection. The Table 5.2 details the outcomes achieved with various optimizers during training and testing phases.

Table 5.2 – MobileNet Ocular Disease with Various Optimisers

OPTIMISER	TRAINING ACCURACY	TESTING ACCURACY
Adagrad	0.8438	0.78
RMSProp	0.8993	0.72
Adam	0.8669	0.44
SGD	0.8889	0.69

These results shed light on the comparative effectiveness of different models and optimizers in the context of the dataset, providing valuable insights into the AI-

driven classification of ocular diseases. The metrics serve as a basis for understanding the strengths and limitations of each model, contributing to the ongoing refinement and optimization of the overall system.

5.3.1 MobileNetV2 Training Progress

This section presents a visual depiction of the training progress of the MobileNetV2 model across 30 epochs. The first graph in the Fig. 5.9 portrays the evolving accuracy of the model, showcasing a continuous improvement trend over successive epochs. Concurrently, the second graph in the Fig. 5.9 elucidates the loss metrics, encompassing both training and validation losses, revealing the model's capacity to optimize and generalize to new data. These graphical representations offer a comprehensive insight into the learning trajectory of MobileNetV2, providing a nuanced understanding of its performance dynamics during the training phase.

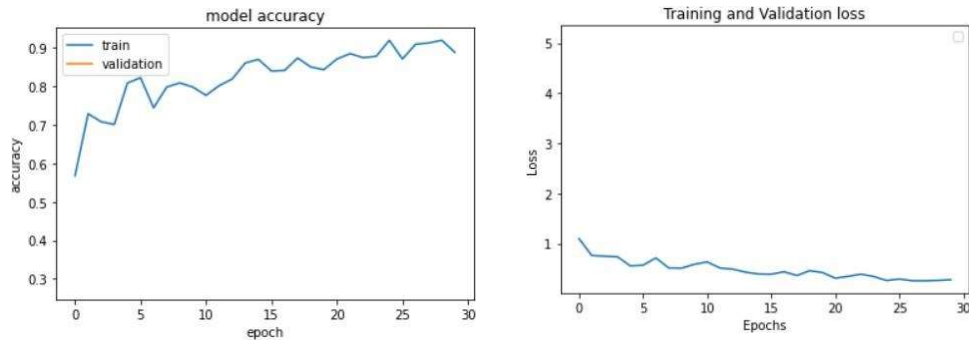


Figure 5.9 – Accuracy Evolution and Loss Metrics across 30 Epochs

5.4 INTERPRETABILITY OF IMAGE RESULTS

This section elucidates the outcomes of the Explainable Artificial Intelligence (XAI) model. In this segment, the input fundus images, alongside the corresponding output generated by the MobileNetV2 model as shown in the Fig. 5.10, are subjected to Local Interpretable Model-agnostic Explanations (LIME). The

resultant images showcase specific highlighted regions, providing interpretability to the AI-driven ocular disease predictions. These visual representations contribute to a more transparent understanding of the decision-making process of the MobileNetV2 model, aiding healthcare professionals in comprehending and verifying the AI-based diagnoses. The XAI results serve as a pivotal component in fostering trust and confidence in the model's predictions within the context of ocular disease management.

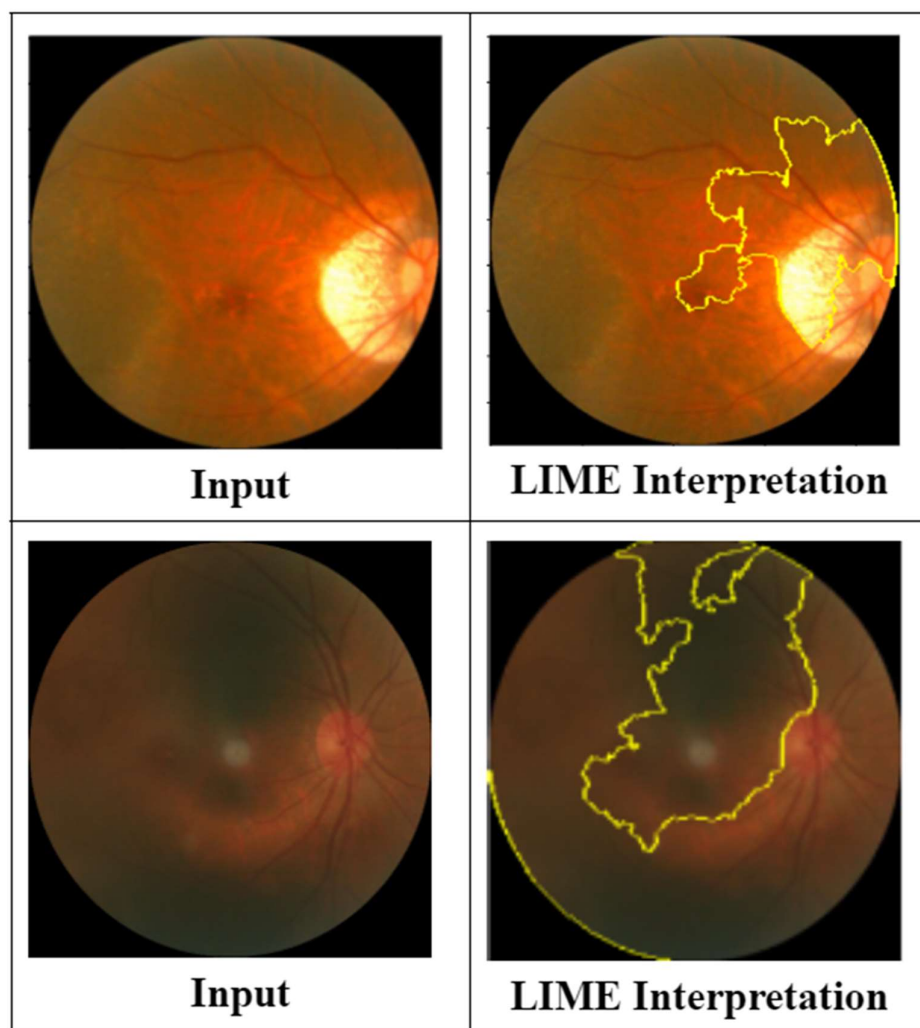


Figure 5.10 – Illustration of XAI Model Results

CHAPTER 6

CONCLUSION AND FUTURE WORK

This section discusses the conclusion and further improvements which can be possibly made in SMC-BC.

6.1 CONCLUSION

In order to make EHR systems interoperable across multiple hospital organizations, doctors can easily track patient medical histories, patients do not need to carry medical history files, and there will be substantial enhancements in digital records. An EHR scenario falls under the private and closed blockchain category, and this solution can effectively conclude that it is an urging framework of this kind of blockchain. It offers a dependable and secure solution in managing medical field records. HLF is an offering blockchain system that has some concepts of policies, smart contracts, and provision of secure personal identities which make the records secure and controlled. Patients can now access the EHR with security and self-service capabilities thanks to the SMC-BC. Additionally, the SMC-BC supplied the fundamentals of FL and XAI to bring about a revolution in EHR data, specifically in Ocular based EHR. An excellent way to guarantee the validity of AI models in healthcare is to conduct experiments on different models using datasets and to overcome dataset scarcity through data augmentation. By granting confidentiality to the data that will be trained when the doctor modifies the AI predicted class, FL's integration into the HLF network enhances the security of the data. The XAI plays a vital role to bring trust on the AI models to patients and as well as to the doctor by explaining the decisions made by the AI models. All these technologies together provide a robust, secured, self-servicing, trustworthy community to manage the EHR which sets a new path in the medical field.

6.2 FUTURE WORK

The SMC-BC has several future scope that could be included to benchmark its advantages in the healthcare sector. Handling healthcare data is crucial and needs more security and privacy to the data being shared. SMC-BC is not compromising on the privacy, it provides phenomenal privacy and security to the patients data. The SMC-BC is a community developed for a use case of ocular disease EHR management that can in future be extended to multiple other use cases where the security and trust of data is a major concern.

In future , in addition to using SMC-BC for other use cases the potential of HLF is showcased which helps in applying HLF based solutions to other applications. The need for secure data training with AI models in today's medical world is important with concern of patient privacy which is achieved by FL and independent docker execution in the network. The potential future development in SMC-BC can be improving User Interface (UI) with low latency and fast prediction with docker enabled XAI explanations.

Other development could be developing a text based prediction considering all other parameters and providing an efficient text-based report based on the decision made with text explainer tools.

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